DIALA LTEIF

AI Scientist | Healthcare Applications | Multimodal AI | Medical Imaging Solutions

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SUMMARY

Experienced AI researcher and software scientist with a PhD in Computer Science focused on healthcare applications. Demonstrated success in leading projects integrating explainable AI and multimodal data for neurodegenerative disease diagnosis. Strong background in designing scalable AI pipelines and collaborating across domain experts to deliver impactful AI tools in medical imaging. Eager to leverage leadership and research experience to drive AI innovation.

EDUCATION

PhD in Computer Science — 09/2019 - 05/2026
Boston University — Boston, MA, USA
GPA | **3.86** / 4.00

B.Sc. in Computer Science, with distinction — 09/2016 - 05/2019
American University of Beirut — Beirut, Lebanon
Major GPA | **4.00** / 4.00

PUBLICATIONS

- Vision-language framework for multi-sequence brain magnetic resonance imaging. MedRxiv — 2026
  Lteif, D., Jia, S., Bit, S., Kaliaev, A., Mian, A. Z., Small, J. E., ... & Kolachalama, V. B.
- Higher-Order Domain Generalization in Magnetic Resonance-Based Assessment of Alzheimer's Disease. arXiv preprint arXiv:2601.01485 (under review). — 2026
  Batool, Z., **Lteif, D.**, Kolachalama, V. B., Ozkan, H., & Aptoula, E.
- Anatomy-Guided, Modality-Agnostic Segmentation of Neuroimaging Abnormalities. Human Brain Mapping 46(14), e70329. — 2025
  Lteif, D., Appapogu, D., Bargal, S. A., Plummer, B. A., & Kolachalama, V. B.
- AI-based differential diagnosis of dementia etiologies on multimodal data. Nature Medicine, 1-13. — 2024
  Xue, C.*, Kowshik, S.*, **Lteif, D.**, Puducheri, S., Zhou, O., Walia, A., ... & Kolachalama, V. B.
- Disease-driven domain generalization for neuroimaging-based assessment of Alzheimer's disease. Human Brain Mapping, 45(8), e26707. — 2024
  Lteif, D., Sreerama, S., Bargal, S. A., Plummer, B. A., Au, R., & Kolachalama, V. B.
- VISDA 2022 challenge: Domain adaptation for industrial waste sorting. In NeurIPS 2022 Competition Track (pp. 104-118). PMLR — 2023
  Bashkirova, D., Mishra, S., **Lteif, D.**, Teterwak, P., Kim, D., Alladkani, F., Akl, J., Calli, B., Bargal, S.A., Saenko, K., Kim, D., Seo, M., Jeon, Y., Choi, D., Ettedgui, S., Giryes, R., Abu-Hussein, S., Xie, B. Li, S.
- Ani-GIFs: A benchmark dataset for domain generalization of action recognition from GIFs. Frontiers in Computer Science, 4. doi:10.3389/fcomp.2022.876846 — 2022
  Majumdar, S. S.*, Jain, S.*, Tourni, I. C.*, Mustafin, A., **Lteif, D.**, Sclaroff, S., Saenko, K., Bargal, S. A.

SKILLS

Artificial Intelligence, Deep Learning, Explainable AI, Computer Vision, Multimodal Large Language Models (MLLMs), Multi-agent Systems, Scalable AI, Image Processing, Medical Imaging, Multi-GPU Training, Statistical Analysis, Interdisciplinary Research

Programming Languages

Python, Java, C, C++, C#, JavaScript, PHP, Matlab, SQL

Relevant Tools

GitHub, HuggingFace, TensorFlow, PyTorch, Keras, Tensorboard, Wandb.ai, AWS Cloud

KEY ACHIEVEMENTS

- **Grace Hopper Celebration Conference Award**
  Boston University 2020
- **Dean's Honor List**
  American University of Beirut 2016-2019
- **First place winning team in the ACM Lebanese Collegiate Programming Contest**
  Beirut, Lebanon 2017
- **Fifth place winning team over Lebanon in the IEEExtreme Programming Competition**
  2017

GRADUATE COURSEWORK

CS585: Image and Video Computing

CS542: Machine Learning

CS640: Artificial Intelligence

CS530: Advanced Algorithms

CS535: Complexity Theory

CS552: Operating Systems

CS511: Formal Methods

EXPERIENCE

Research Fellow
01/2020 - Present

Boston University — Boston, Massachusetts, USA

- Developing a large-scale vision–language framework for multimodal brain MRI interpretation with multi-agent systems, privacy-aware workflows, and clinician evaluation (*in progress*).
- Designed anatomy-guided approaches for segmenting neuroimaging abnormalities outperforming state-of-the-art brain tumor segmentation methods *(HBM '25*).
- Integrated anatomy and interpretability in the training of generalizable models for the assessment of Alzheimer's Disease (*HBM '24*).
- Contributed to developing a multimodal AI tool for the differential diagnosis of dementia etiologies—built imaging-model pipeline and ran training/validation workflows (*Nature Medicine '24*).
- Targeted domain generalization and explainable AI on benchmark natural imaging datasets (*Frontiers in CS, '22*).
- Experimented with cross-layer parameter sharing for knowledge transfer and resource-efficient on-device machine learning *(2020-2021)*.

Applied AI/ML Scientist Intern
06/2022 - 09/2022

Inari Medical Inc. — Irvine, CA, USA

Developed AI solutions for ultrasound devices to enhance endovascular procedures.

Software Development Intern
08/2018 - 09/2018

PinPay SAL, Beirut Digital District — Beirut, Lebanon

Worked on Amazon Alexa, Google Assistant services and front-end web development.

Teaching Fellow
09/2019 - Present

Boston University — Boston, Massachusetts, USA

CS511: Formal Methods I (Fall 2025)
CS585: Image and Video Computing (Spring 2025)
CS542: Machine Learning (Fall 2021)
CS330: Introduction to Algorithms (Spring 2020)
CS103: Introduction to Internet Technologies and Web Programming (Fall 2019)

Instructor
07/2018

Future Developer Summer Camp, American University of Beirut — Beirut, Lebanon

Taught Unity and iOS development to students between the ages 12 and 18

PROFESSIONAL ACTIVITIES

Reviewer
2024 - Present

IJCV, Frontiers in Neuroimaging, CVPR, CAI, Alzheimer's & Dementia Journal

Seminar Co-chair
09/2021 - 06/2022

BU Image and Video Computing Group — Boston University

Invite speakers and organize weekly seminars at the IVC group at Boston University.

Guest Speaker
08/2021

BU AI4ALL Program — Boston University

Gave a talk to high school students about Deep Learning and Explainable AI.

Graduate Student Advisor
2020 - 2021

BU ACM-W Student Chapter — Boston University

Coordinated the student mentorship program and advocated for full engagement of women in computing and technology.

Competitive Programming Contestant
11/2017

ACM Arab Collegiate Programming Competition 2017 — Egypt

Teamed with fellow AUB students to compete in the 9th Arab Collegiate Programming Competition and ended up in the 31st place in the MENA region out of 104 participating teams.

SELECTED COURSE PROJECTS

CS585 Final Project: Image Deblurring Applied to Recycling Data
2021 — Boston University

Applied and evaluated existing classical and deep learning deblurring methods to a video of recycling trash moving on a conveyor belt.

CS542 Final Project: Image Classification on Covid-19 X-rays
2020 — Boston University

Implemented binary and multi-class classification using VGG and ResNet backbone architectures in Keras on a Covid-19 X-ray dataset.

CS640 Final Project: Facial Expression Analysis Based on Videos of Presidential Candidates
2019 — Boston University

Worked with classmates to extend existing state-of-the-art deep convolutional neural networks for sentiment analysis on video clips of the 2020 Democratic Presidential candidates.

Undergraduate Final Year Project: Emotion Recognition in an Uncontrolled Environment
2019 — American University of Beirut

Created an Android Application that predicted users' emotional state based on collected readings from ShimmerV3 Biophysical sensors and user self-assessment (in collaboration with the Psychiatry Department at AUBMC).